

Neonatal Jaundice

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Neonatal Jaundice

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Introduction

Neonatal jaundice is the yellowish discoloration of the **eyes, skin and mucous membranes in the first month of life**, due to an elevated level of bilirubin in the blood. It is common and in most cases a harmless problem in neonates.

It may present with other symptoms:

- Yellowing of the palms
- Dark, yellow urine — a newborn's urine should be colourless
- Pale-coloured stool — should be yellow or orange

The yellow discoloration may result from elevation of **either the unconjugated or the conjugated** bilirubin.

- Bilirubin may have a physiologic role as an antioxidant, but elevations of **indirect, unconjugated** bilirubin are potentially **neurotoxic**.
- The conjugated form is **not neurotoxic**, but direct hyperbilirubinaemia indicates a potentially serious hepatic disorder or systemic illness.
- Poor management can result in either death or serious neurological problems.

Epidemiology

- First described in a child textbook 1000 years ago.
- In 1875 **Orth** first described yellow staining of the brain, a pattern later referred to by **Schmorl** as kernicterus.
- Extremely common: almost every newborn develops an unconjugated serum bilirubin $> 30 \text{ } \mu\text{mol/L}$ (1.8 mg/dL) during the first week of life.
- Incidence varies with ethnicity and geography — higher in East Asians and American Indians, **lower in Africans**.
- Incidence is higher in populations living at **high altitudes**.
- Risk of significant neonatal jaundice is higher in **male** infants.
- Prevalence in hospital practice is estimated at **50–80%**: about **80% in preterm**, about **50% in term** infants.

Aetiology

The cause of jaundice can be either physiologic or pathologic.

Causes of physiological jaundice

1. **Decreased RBC survival** (< 90 days), increased RBC volume/kg, polycythaemia of the newborn
2. **Poor hepatic uptake** due to immature liver — decreased ligandin or Y-protein
3. **Increased enterohepatic circulation** due to:
 - High level of intestinal beta-glucuronidase
 - Delayed colonisation by bacteria
 - Decreased gut motility

Causes of pathological jaundice

Divided into **unconjugated** (haemolytic / non-haemolytic) and **conjugated** (hepatic / post-hepatic).

Haemolytic, unconjugated — by Coombs test:

Coombs test positive	Coombs test negative
ABO incompatibility	RBC membrane defect (spherocytosis, elliptocytosis)
Rh incompatibility	RBC enzyme defect (G6PD)
	Haemoglobinopathies
	Sepsis
	DIC
	Haematomas
	Polycythaemia

Drugs (diazepam, oxytocin) are also a haemolytic cause.

Non-haemolytic, unconjugated:

- Breast milk jaundice
- Crigler–Najjar syndrome types I and II
- Gilbert syndrome
- Hypothyroidism

Conjugated — hepatic:

- Idiopathic neonatal hepatitis
- Dubin–Johnson syndrome
- Rotor’s syndrome
- Infections — TORCH, sepsis
- Inborn errors of metabolism (galactosaemia, tyrosinaemia)

Conjugated — post-hepatic:

- Biliary atresia
- Bile duct stenosis
- Choledochal cyst

Common causes in our environment

- Physiological
- Blood group incompatibility
- G6PD deficiency
- Breast milk jaundice
- Sepsis
- Cephalhaematoma

Clinical presentation

- Yellowness of the eyes or body
- Fever
- Vomiting
- Inability to suck
- Features of failure to thrive
- Abdominal distension
- Passage of pale stool

History

- Biadata
- Onset / duration / progression
- Colour of urine and stool
- Vomiting
- Feeding history
- Loss of weight
- **Maternal history** — antenatal booking, illness, infections
- **Drugs** — sulfa, NSAIDs, herbal
- **Medical history** — blood group, haemolytic diseases, liver disease

- **Birth history** — trauma, delayed cord clamping

Examination

Examine for jaundice — **bright yellow/orange** or **greenish/muddy yellow**. Jaundice becomes apparent in a **cephalocaudal progression**, starting on the face and progressing to the abdomen and then the feet as serum levels increase.

Kramer's Index

Zone	Area	Approximate bilirubin
1	Face	4–6 mg/dL
2	Chest & upper trunk	8–10 mg/dL
3	Lower abdomen, thigh	12–14 mg/dL
4	Forearms & lower legs	15–18 mg/dL
5	Palms & soles	> 15–20 mg/dL

Also examine for:

- Pallor
- Dehydration
- Weight loss
- Fever or hypothermia
- Hepato/splenomegaly
- Evidence of birth trauma or injuries
- Evidence of kernicterus

Investigations

Bilirubin measurement

- Serum bilirubin (total and direct)
- Transcutaneous (bilirubinometer / icterometer)

Others

- Full blood count and blood smear
- Haemoglobin concentration
- Reticulocyte count
- Blood group
- Coombs test
- Urinalysis — reducing substances
- Urine m/c/s
- Electrolytes, urea and creatinine
- Liver function tests (AST/ALT, ALP/GGT)
- G6PD assay
- Ultrasound scan
- Thyroid scan and thyroid function tests

Transcutaneous measurement of bilirubin (TcB)

Eligible:

1. Babies ≥ 38 weeks gestation, visibly jaundiced after 24 hours of age until 14 days
2. Babies 34–37+6 weeks gestation who are ≥ 72 hours old, until 14 days
3. Babies **not** on phototherapy

Contraindications:

1. Babies with pathological jaundice
2. Babies less than 24 hours old
3. Babies on phototherapy, or who have undergone phototherapy / exchange transfusion

Jaundice associated with breast-feeding

	Breast milk jaundice	Breastfeeding jaundice
Frequency	2% of breastfed infants	13% of breastfed infants
Onset	After the 7th day	Within 1 week
Peak level	Up to 10–30 mg/dL; peaks at 2–3 weeks	May reach > 12 mg/dL
Course	SB falls gradually if breastfeeding stops	—
Response	Responds to phototherapy	Responds to frequent breastfeeding
Aetiology	Not clear; associated with glucuronidase in milk	Decreased milk intake, dehydration, reduced caloric intake, glucose water supplement

Treatment of unconjugated hyperbilirubinaemia

Aims

1. Reduce the incidence of severe hyperbilirubinaemia
2. Prevent bilirubin encephalopathy (kernicterus)

Modalities: phototherapy, exchange blood transfusion, pharmacotherapy.

Phototherapy (PT)

The use of high intensity light energy to reduce bilirubin levels on the skin surface.

- Light is used in the **white, blue, turquoise and green** wavelengths
- Bilirubin absorbs light maximally in the **blue range, 420–470 nm**
- Useful in prevention and/or treatment of moderate hyperbilirubinaemia
- When adequately delivered, should lower bilirubin by **1–2 mg/dL over 4–6 hours**

Mechanism of action

Conversion of insoluble bilirubin into soluble bilirubin, by:

- **Structural isomerization** — conversion to lumirubin, rapidly excreted in bile and urine
- **Photoisomerisation** — converts bilirubin (4Z,15Z) to soluble isomers (e.g. 4Z,15E) without conjugation, excreted into bile
- **Photooxidation**

Indications

- TSB > 15 mg/dL in term
- TSB > 12 mg/dL in preterm
- TSB > 5 mg/dL within 24 hours
- Adjuvant to exchange transfusion
- Prophylactic PT — ELBW, bruised babies, haemolytic disease of the newborn, VLBW with perinatal risk factors

Contraindications

- Porphyria
- Conjugated hyperbilirubinaemia — bronze baby syndrome
- Unconjugated hyperbilirubinaemia 20 mg/dL or greater
- Diseases with hypersensitivity to light

Procedure / precautions

- Distance from skin — 45 cm; intensive PT — 15–20 cm
- Space of 5–8 cm between phototherapy unit and incubator
- Double surface PT can be given by fibre-optic blankets (biliblankets)
- Distance the child from others not needing PT
- Cover the eyes and genitals
- Change position every 2–4 hours
- Level to be checked every 10–20 hours
- Frequent temperature monitoring
- Supplemental hydration (10–25 mL/kg/day)
- Daily weight check
- Watch for side effects

Complications

Early	Late
Loose stools	Skin malignancy
Dehydration	Damage to DNA
Skin rashes	Patent ductus arteriosus
Hyperpigmentation	Gonadal damage
Upsets maternal–infant bonding	Retinal damage
Hyperthermia	Disturbance of circadian rhythm
Bronze baby syndrome	

Exchange blood transfusion (EBT)

Done to rapidly lower the serum bilirubin concentration and/or prevent further rise.

Indications

- Unconjugated hyperbilirubinaemia ≥ 20 mg/dL in term, or ≥ 15 mg/dL in LBW preterm
- Serum bilirubin up to 10 mg/dL for other preterms
- Bilirubin rising rapidly, up to 5 mg/dL/day
- Serum bilirubin > 10 mg/dL on the first day of life, or 15 mg/dL at 48 hours
- Clinical signs of kernicterus

- Others — severe anaemia, DIC

Technique

- Commonly the “**push–pull**” method using an umbilical vein catheter
- Choice of blood — **heparinised fresh whole blood preferred**; ACD or CPD can also be used
- Quantity — **twice the blood volume** of the patient (160–170 mL/kg), over 60–90 minutes
- Blood exchanged in aliquots of 5, 10 or 20 mL, or smaller volumes for small infants
- Priming with **1 mg/kg salt-poor albumin** a few hours before EBT increases efficiency

Procedure

1. Take vital signs
2. Empty the stomach — prevents aspiration
3. Ensure the right blood is available
4. Maintain asepsis
5. Catheterise the umbilical vein and secure in place
6. Take pre-EBT PCV and serum bilirubin
7. Exchange blood in aliquots and document
8. Give 1 mL of 10% calcium gluconate after every 100 mL slowly, listening to the heart alongside
9. Take post-EBT PCV and serum bilirubin
10. Replace all drugs given earlier if not time for the next dose
11. Prevent hypoglycaemia — 200 mg/kg of IV glucose
12. Adjuvant phototherapy

Complications

Of catheterisation: gut perforation and peritonitis; wrong position of catheter tip (may excite cardiac arrhythmia if in the right atrium); necrotising enterocolitis; liver cirrhosis (long-term).

Of blood transfusion: transmission of infection (malaria, hepatitis B, HIV); transfusion reactions (immediate, delayed); thromboembolism.

Of the EBT itself: haemodynamic changes from the push–pull process (hypovolaemia, hypervolaemia); electrolyte changes (hyperkalaemia, hypocalcaemia, hypomagnesaemia); acid/base imbalance; changes in cerebral blood flow.

Pharmacotherapy

Not routine, because of ineffectiveness.

- **Phenobarbitone** — augments hepatic uptake of bilirubin and increases activity of glucuronyl transferase (UDPG-T). Takes 48–72 hours to reach therapeutic level. Dose 5–8 mg/kg/day. Side effect: sedates, making the baby difficult to assess. Useful in Crigler–Najjar disease.
- **Metalloprotoporphyrins** (e.g. tin mesoporphyrin) — competitive inhibitors of haem oxygenase. Experimental, showing prospects.
- **Oral agar** (e.g. cholestyramine) — decreases enterohepatic circulation.
- **Albumin infusion** — increases albumin binding to bilirubin for easy transport.
- **Intravenous immunoglobulin** — for infants with Rh or ABO isoimmunisation; inhibits haemolysis. Dose 0.5–1 g/kg/dose IV, repeat in 12 hours.

Kernicterus

The **pathologic** diagnosis referring to yellow staining of the brain caused by bilirubin deposition, associated with neuronal necrosis and gliosis. It is a direct consequence of very high levels of unconjugated bilirubin.

Parts of the brain commonly affected

- Basal ganglia
- Brain stem nuclei
- Cerebellar nuclei
- Hippocampus
- Anterior horn cells of the spinal cord

By pathologic criteria it develops in **30%** of infants with untreated haemolytic disease and bilirubin levels $> 25\text{--}30$ mg/dL. Overt neurologic signs carry a grave prognosis: **more than 75% of infants die**, and **80% of affected survivors** have bilateral choreoathetosis with involuntary muscle spasms.

It manifests as acute or chronic encephalopathy. The acute phase has **3 phases**:

- **Phase 1 (days 1–3):** hypotonia, lethargy, high-pitched cry, poor suck, loss of Moro reflex
- **Phase 2 (middle of first week):** hypertonia, opisthotonus, rigidity, oculogyric crisis, retrocollis, fever, seizures. Survivors develop chronic bilirubin encephalopathy
- **Phase 3 (after first week):** hypotonia replaces hypertonia

Chronic encephalopathy

First year: hypotonia, active deep tendon reflexes, obligatory tonic neck reflexes, delayed motor skills.

After the first year: movement disorders (choreoathetosis, ballismus, tremor), upward gaze, sensorineural hearing loss.

Treatment of kernicterus

- Counselling / training the parents on coping with the sequelae
- Management of feeding difficulty
- Physiotherapy for limitation of disability
- Speech and language therapy (SALT)
- Follow-up appointments

Treatment of conjugated hyperbilirubinaemia

Mainly supportive:

- Parenteral vitamin K
- Vitamin A and D supplementation — prevents rickets
- Diet — low fat, medium chain triglyceride
- Phenobarbitone — helps bile excretion
- Surgery may be needed (not usually in the neonatal period)

Prevention

Follows the **5 levels of prevention**:

1. **General health promotion** — educate the masses about it and how to prevent it; lifestyle modification

2. **Specific protection** — identify risks and institute preventive measures
3. **Early diagnosis and adequate treatment** — identify and treat jaundice promptly
4. **Limitation of disability** — physiotherapy
5. **Rehabilitation** — integration of the baby and the mother into the community

Conclusion

Neonatal jaundice still contributes significantly to neonatal morbidity, mortality and long-term handicap, especially in developing countries. This is made worse by ignorance on the part of parents and health workers, which contributes to delay in seeking proper medical care.

Education at primary health care level, prompt diagnosis and referral to the appropriate specialist will go a long way to preventing jaundice-associated problems early, so that death is averted and the child has a good quality of life.